A Visual Trace Map of Edinburgh Place Names in Literature

Merritt Wang

Master of Science
School of Informatics
University of Edinburgh
2026

Abstract

Existing digital tools for literary geography plot place names as points on a map, treating them as geographic data. The meaning a place name carries in a literary text, however, comes from its narrative context rather than its geographic coordinates. This dissertation asks whether that narrative structure can be made visible directly, using the LitLong Edinburgh database of 620 literary works (1687–2015), 2,135 place names, and 50,248 place-name mention records. Two research questions are addressed: RQ1 asks how an author's distribution of place-name mentions across the city can be visualised as a distinctive spatial "fingerprint," and RQ2 asks how the co-occurrence of place names within literary works can be visualised to reveal a narrative topology distinct from their geographic topology. Six interactive D3.js visualisation prototypes were designed and implemented across these two directions: a radar chart, a bar-code fingerprint, and small-multiple maps for RQ1, and a force-directed network, a linear connection diagram, and a metro-style map for RQ2, all evaluated through a user study with domain-expert and general-public participants. The corpus analysis underlying RQ2 shows that Leith and Princes Street, roughly 2.5km apart, are the most strongly co-occurring place pair in the corpus (28 shared works). The user study ($n = 13$) tested whether participants could match blind, unlabelled fingerprints back to their authors. Results support a qualified yes for RQ1: the radar chart was matched to its correct author 69% of the time (against a 20% chance baseline for five authors), with the bar-code and small-multiples designs close to half correct, and the radar chart the most consistently distinguishable design across every measure used, though single-shape identification accuracy was more modest and varied considerably by design.

Research Ethics Approval

This project obtained approval from the Informatics Research Ethics committee.
Ethics application number: 446635
Date when approval was obtained: 13 May 2026
The participants' information sheet and a consent form are included in the appendix.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(*Merritt Wang*)

Acknowledgements

I would like to thank my supervisor, Uta Hinrichs, for her guidance and feedback throughout this project. I am also grateful to the School of Informatics at the University of Edinburgh for the resources and support that made this dissertation possible, and to everyone who took part in the user study.

Table of Contents

List of Figures

Chapter 1

Introduction

Edinburgh, the first UNESCO City of Literature, has an unusually rich corpus of literary texts set in or concerning the city. The Palimpsest project at the University of Edinburgh systematically extracted place-name mentions from hundreds of these works spanning three centuries, producing the LitLong Edinburgh database: 620 documents published between 1687 and 2015, 2,135 distinct place names, and 50,248 geo-referenced mention records [1]. The project made this corpus publicly accessible through a web interface that plots place mentions on an interactive geographic map.

Geographic maps are a natural starting point for literary data of this kind, but they carry an implicit assumption: that the meaning of a place name is determined by its location. That assumption holds up fine in geography, less so in literature. When Irvine Welsh names Leith in *Trainspotting* (1993), the word is not primarily a set of coordinates north of Edinburgh city centre; it evokes a social world, a class position, a political history. When Walter Scott names the Canongate in *The Heart of Midlothian* (1818), he is invoking a specific stratum of Edinburgh's past. A geographic map can show where these places are, but it has nothing to say about when in the narrative they appear, how often they recur, which other places surround them, or whose authorial voice chose them.

Place names in literary text, in other words, carry meaning beyond their geographic coordinates — meaning bound up with a work's social and political landscape, and made legible through narrative structure: how often a place recurs, which other places surround it, and where in the story it appears. Accordingly, the visualisation problem tackled here is less about rendering literary geography accurately on a map, but more about representing narrative structure in a form that reveals patterns geographic tools cannot.

1.1 Problem Statement

This dissertation addresses two core questions: how can place-name mentions in literature be visualised so that they can be read as distinctive authorial signatures? And how can visualisation help compare place-name-related narrative links and topologies across literary works?

It divides into two sub-problems corresponding to two distinct dimensions of narrative structure. The first concerns *authorial spatial attention*: does each author's distribution of place-name mentions across Edinburgh constitute a distinctive pattern that can be recognised without prior knowledge of the author? The second concerns *narrative co-occurrence*: do places that appear together across literary works form a topology that differs systematically from their geographic arrangement?

Existing digital literary-geography projects, such as Stange and Dörk's Novel City Maps and Piatti et al.'s Literary Atlas of Europe, do not address either question: they do not separate literary space from geographic space, do not support comparison of authorial spatial styles, and do not reveal which places are narratively proximate, as opposed to geographically proximate.

1.2 Research Questions

Two formal research questions guide the project:

RQ1 (Author Spatial Fingerprint): Does each author's distribution of place-name mentions across Edinburgh's official neighbourhood sectors form a visually distinctive spatial signature, sufficient for users to identify authors from their fingerprint alone? Success requires different authors to produce visually distinct shapes on a one-to-one basis, with no two shapes indistinguishable to a naive user.

RQ2 (Narrative Topology): Does the co-occurrence of place names within literary works reveal a narrative topology that diverges meaningfully from the geographic topology of Edinburgh? Success requires the visualisation to surface cases where narrative proximity contradicts geographic proximity — places that are spatially distant yet repeatedly appear together in the same works.

1.3 Methods

To answer these questions, six interactive D3.js visualisation prototypes were designed and implemented across the two directions (three per RQ), and evaluated through a user study comparing domain-expert and general-public responses.

1.4 Contributions

This dissertation makes the following contributions:

1. A visualisation case study of the LitLong Edinburgh corpus [1], applying two complementary directions – author spatial fingerprints (three designs, RQ1) and narrative topology networks (three designs, RQ2) – implemented in D3.js as self-contained, deployable HTML files, available at `https://merrittw-1307.github.io/uoe-edinburgh-literary-geovis/`.
2. Empirical evidence from a user study ($n = 13$) that a spatial-fingerprint encoding is learnable: participants correctly matched unlabelled radar-chart fingerprints back to their authors 69% of the time (against a 20% chance baseline), with the radar chart the most consistently distinguishable of the three fingerprint designs tested.
3. A spatial classification of Edinburgh place names derived from official City of Edinburgh Council boundary data: 14 sectors from the Neighbourhood Partnership Areas and Natural Neighbourhoods datasets (Open Government Licence v3.0).
4. A method for extracting narrative reading order from LitLong via `start_word`, strictly monotonic within each document, enabling position-normalised analysis of when and where each place appears.

1.5 Dissertation Overview

Chapter 2 situates the project within literary geography and geocriticism, existing literary-corpus visualisation, and information visualisation principles for humanities data. Chapter 4 covers data exploration and the design decisions for each of the six visualisation candidates. Chapter 5 covers the technical implementation, from database setup to the D3.js prototypes. Chapter 6 describes the user study and reports its

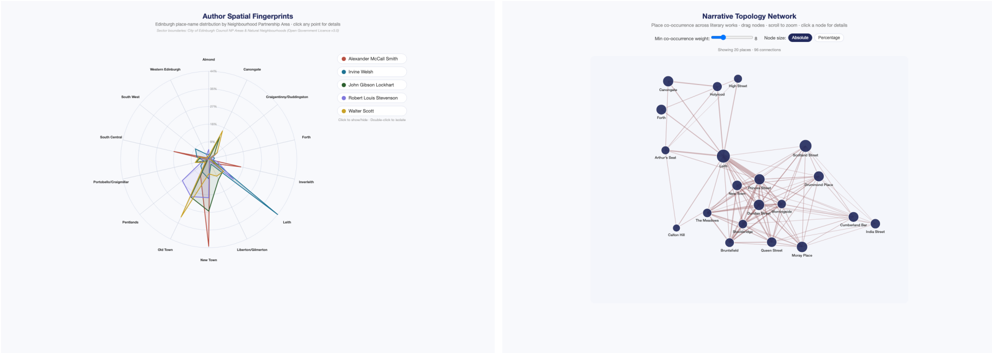

Figure 1.1: The primary design for each direction: the radar chart (RQ1) and the force-directed network (RQ2).

results. Chapter 7 discusses the findings against the research questions and the project's limitations. Chapter 8 concludes and outlines future work.

Chapter 2

Literature Review

This chapter situates the project within four bodies of literature: theoretical work on literary geography and geocriticism, the empirical foundation provided by the LitLong Edinburgh project, existing approaches to visualising place in literary corpora, and information visualisation principles relevant to humanities data.

2.1 Literary Geography and Geocriticism

The relationship between literature and place has been a concern of literary scholarship since at least the late twentieth century, but the theoretical grounding for a spatially-oriented reading of fiction consolidated around what is now called the "spatial turn" in the humanities [17]. Bertrand Westphal's geocriticism argues that literary places are not simple reflections of real geographies but active constructions: representations that reshape the reader's understanding of space and that exist in dialogue with the physical world rather than merely depicting it [19]. On this account, the place names in a novel are not pointers to locations on a map but nodes in a narrative network.

This theoretical consolidation was accompanied by a parallel empirical strand: the edited volume *The Spatial Humanities* surveys how geographic information systems (GIS) can be adapted to represent spatial relationships in humanities texts [3], and Cooper and Gregory's literary GIS of Lake District travel writing demonstrates that such systems can encode the qualitative, subjective spatial experience recorded in a text rather than only its objective coordinates [6]. Both works establish that a GIS-adjacent representation of literary place need not default to strict geographic accuracy, a premise this project extends beyond mapped coordinates entirely.

Franco Moretti's quantitative approach to literary geography offered a different perspective [11]. By mapping the settings of hundreds of European novels, Moretti demonstrated that literature does not distribute itself uniformly across geographic space: certain places are obsessively returned to, others systematically avoided. He coined the phrase "literary silences" for the latter, absences that turn out to be as informative as presences. Moretti's maps are geographic, but his interpretive framing is already narrative: he reads them for patterns of attention and neglect, not spatial accuracy.

Both traditions converge on a point that motivates this project: place in fiction is a narrative construction rather than a geographic fact. The visualisation problem, then, is less about representing the geography of literary Edinburgh more faithfully, and more about making its narrative dimensions (frequency, authorial style, co-occurrence, temporal position) visible.

2.2 The LitLong Edinburgh Project

The empirical foundation of this project is the LitLong Edinburgh corpus, produced by the Palimpsest project at the University of Edinburgh [1], which used NLP pipelines to extract geo-referenced place-name mentions from literary works set in or concerning Edinburgh (1687–2015). The full database, accessed here via a 2022 PostgreSQL dump, contains 620 documents, 2,135 place names, 50,248 mention records, and 424 authors across 29 genre classifications; the specific database fields this project draws on are described in Chapter 5. The original LitLong web interface renders mentions as dots on an interactive geographic map – intuitive and technically accomplished, but treating geographic location as the primary axis of literary meaning. This project treats the same underlying data differently: as raw material for representing narrative structure rather than geographic position.

2.3 Visualising Place in Literary Corpora

The closest antecedent to the present project is the work of Stange and Dörk on visualising literary place mentions in novels set in Berlin and New York [16]. Their "Novel City Maps" approach generated fingerprint-like representations of each novel's spatial distribution across the city, using street-map coordinates as the positional axis. The present work extends this in a significant direction, replacing geographic coordinates entirely with an abstract categorical axis (city sectors), so the fingerprint shape represents authorial spatial attention without the perceptual noise real-world topography introduces. The data drove this choice: as Section 4.2 reports, place names in the LitLong corpus are too sparsely shared across authors for individual place names to serve as meaningful axes.

Other digital literary geography projects have largely retained the geographic map as their primary interface. *A Literary Atlas of Europe*, an ETH Zurich cartography project mapping the fictional geographies of European literature across several test regions, plots textual settings against real-world coordinates using specialised uncertainty-visualisation techniques for vaguely or fictionally located places [14]. Valuable as such tools are for exploring the geographic distribution of literary references, they cannot address questions about authorial style, narrative co-occurrence, or the divergence between spatial and narrative proximity.

The information visualisation literature offers relevant precedents for the two directions pursued here. Tufte's small multiples principle [18] (presenting multiple instances of the same visual form side by side for comparison) underlies the Small Multiples design in Direction 1. The force-directed graph layout used in Direction 2 follows Fruchterman and Reingold's energy-minimisation algorithm [8]. The metro-style map adapts Beck's insight that functional topology, the traveller's need to know which line and which direction, can be more usefully represented than geographic accuracy [2].

2.4 Information Visualisation for Humanities Data

Shneiderman's information-seeking mantra, "overview first, zoom and filter, details on demand" [15], provides a structural principle for the interaction design in this project. All six visualisation prototypes follow this pattern: an overview of the full dataset (all authors, all co-occurrence pairs) is presented by default, with interactive mechanisms for filtering and zooming, and detail panels that surface place-specific and book-specific information on demand.

Drucker argues that humanistic data should be understood as *capta*, actively interpreted and constructed, rather than as objectively given [7]. Several design decisions documented in Chapter 4 bear this out: the choice of sectors as axes reflects an interpretive act (what counts as a meaningful spatial unit for literary analysis?), and the choice of document-level co-occurrence as the granularity for the topology network reflects a judgment about what constitutes meaningful narrative co-presence.

Tufte's principle of small multiples [18] provides the theoretical foundation for the side-by-side comparison panel in Direction 1, where five maps of the same spatial extent but different author-specific bubble distributions enable direct visual comparison of authorial spatial patterns.

The overall design process follows Munzner's nested model for visualisation design and validation [12], which decomposes the design task into four cascading levels: domain problem characterisation, data and operation abstraction, encoding and interaction idiom design, and algorithm implementation, each validated by different means. Chapter 4 corresponds to the abstraction and idiom-design levels, and Chapter 6 validates the resulting idioms empirically through the user study, following the model's recommendation that idiom choices be justified against measured task performance rather than designer intuition alone.

2.5 Graph and Network Visualisation

The Fruchterman-Reingold algorithm [8] models graph layout as a physical simulation, with nodes repelling and edges acting as springs – a standard technique for laying out network diagrams without reference to any external coordinate system. N"ollenburg and Wolff's work on algorithmic metro-map layout [13] establishes that schematic map drawing, placing nodes on an octilinear grid while preserving topological correctness and legibility, is computationally hard in the general case.

2.6 Gap and Contribution

The literature reviewed above identifies three gaps that this project addresses. First, while literary geography theory has established that place in fiction is a narrative construction, existing digital tools continue to render literary place names as geographic data. Second, no existing tool supports direct visual comparison of authorial spatial styles as abstract fingerprints. Third, no existing tool visualises the co-occurrence topology of literary place names as a structure distinct from geographic proximity. This project addresses all three gaps through a system of interactive visualisations grounded in real corpus data and evaluated through a user study.

Chapter 3

Data and Data Preparation

3.1 Data Pipeline

Database setup

PostgreSQL 16 was installed via Homebrew and a `litlong_edinburgh` database created; the full LitLong dump (145 MB, ≈2.3M lines) was imported directly via `psql`. This dump is more complete than the four CSV snapshots provided with the project brief (620 documents versus 548, 50,248 mention records versus 48,056, and 27 additional tables including `api_sentence` and `api_genre`). One table, `api_location`, failed to import because its PostGIS geometry fields (`geom`, `poly`) require an extension that was not installed; it was instead imported from the CSV snapshot via Python/pandas, keeping only `id`, `text`, `lat`, `lon`, `gazref`, and `ptype`, since none of the visualisation directions need the geometry fields.

Narrative order extraction

The `api_locationmention` table's `start_word` field encodes a global word index per mention, validated as strictly monotonically increasing within each document, so it can reconstruct reading order without parsing the source text. A custom table `mention_order` was built with `word_num` (extracted from `start_word`), `mention_order` (rank within document), and `position_pct` (normalised position, $(\text{word\_num} - \text{min})/(\text{max} - \text{min})$ per document), enabling Reader-plot style timeline visualisation as future work.

Sector classification

The classification procedure is described in Section 4.4, implemented via the `shapely` library for point-in-polygon intersection against two GeoJSON boundary files from the City of Edinburgh Council's ArcGIS REST API (Neighbourhood Partnership Areas and Natural Neighbourhoods). Each place with a valid latitude/longitude was tested against the 14 sector polygons in order: an exact match if inside one, nearest-polygon assignment by Euclidean distance if within the bounding box but in an inter-polygon gap, and exclusion to a catch-all "Outer Scotland" category if outside the box entirely. Results were written to `location_sectors` and exported as CSV.

Co-occurrence computation

Document-level co-occurrence was computed in Python via `itertools.combinations`: for each document, every pair within its set of distinct places had its co-occurrence count incremented, filtered to weight ≥ 2, yielding 403 pairs, exported as `network_edges.csv` and `network_nodes.csv`.

Sentence and book enrichment

A second pass joins `api_sentence` and `api_document` to attach, for every place, a small sample of full source sentences (deduplicated, truncated to 220 characters, with book and author) and a ranked list of books that mention it; for Direction 2, the same join is computed per co-occurrence edge, so the places behind an arc or edge can be traced to the specific books where both appear. This is a separate lookup table keyed by place (and place pair), embedded alongside the existing data, and does not alter any previously computed percentage, count, or weight. Because sampling from `api_sentence` surfaces source text verbatim, it can include informal register or strong language where the original does (e.g. Welsh's Scots-dialect prose) – an authentic property of the corpus rather than a sampling artefact, but worth flagging before a live demonstration.

Chapter 4

Design

This chapter describes the exploratory data analysis that shaped the design decisions, the spatial classification method underlying Direction 1, and the six visualisation designs developed across both research directions.

4.1 Design Rationale

The overarching design philosophy adopted for this project is that narrative structure should take precedence over geographic accuracy: place names are treated as data about how a narrative uses space, not as coordinates to be plotted. This motivated abandoning literal geographic position wherever possible, in favour of abstract encodings that make narrative patterns such as authorial spatial attention and place co-occurrence directly visible.

The choice between static and interactive visualisation was settled early in the project. The LitLong corpus contains 620 documents, 2,135 place names, and 424 authors. A static visualisation that attempted to show all of this simultaneously would be unreadable. Interactivity is necessary to support the "overview first, zoom and filter, details on demand" pattern [15] that allows a reader to form a general impression and then drill into specific authors, sectors, or co-occurrence pairs.

4.2 Data Exploration and Key Findings

Before designing the visualisations, the LitLong data was characterised through a series of exploratory analyses, and the results directly informed the space of possible visualisations. These analyses produced five findings that shaped the design decisions documented below.

Finding 1: Author spatial languages are highly individual. The five test authors (Alexander McCall Smith, Irvine Welsh, John Gibson Lockhart, Walter Scott, and Robert Louis Stevenson) were selected for contrast in mention count, number of works, and thematic range (Appendix B, Author Selection). Among them, no single place name is mentioned by three or more authors, and only 18 are shared by any two – a pattern later confirmed to hold at full corpus scale (Section 7.4), where individual vocabularies overlap only sparsely across all 408 authors with location data. This has a direct design implication: individual place names cannot serve as a shared axis system, since most axes would be zero for most authors; city sectors, by contrast, provide a small, fixed set of axes any author's mentions can be meaningfully grouped into, which motivated their use as the radar chart axes.

Finding 2: Sector distributions align with established literary knowledge. The sector-based analysis (the 14-sector framework is defined in full in Section 4.4) matches established literary-geographic knowledge for all five authors: of McCall Smith's total place-name mentions, 43.6% fall within New Town, consistent with his Scotland Street series; of Welsh's, 44.3% fall within Leith, consistent with *Trainspotting*; of Scott's, 32.0% fall within Old Town, consistent with his historical fiction; and Stevenson shows a notable Pentlands component (17.0% of his mentions), consistent with his documented connection to Swanston – reasonable confidence in the sector classification methodology.

Finding 3: Co-occurrence granularity requires document-level analysis. Sentence-level co-occurrence cannot be computed directly from the LitLong corpus: the `api_sentence` table stores one row per location mention rather than one row per genuine grammatical sentence, so sentences containing multiple place names appear as duplicated rows rather than a single row with multiple mentions, making sentence-level grouping structurally unable to recover same-sentence co-occurrence. Page-level co-occurrence, by contrast, yields up to 92 co-occurring places on a single page, producing graphs too dense to be readable. Document-level co-occurrence, counting pairs of places that appear in the same work, yields 403 pairs with weight ≥ 2 (where weight counts the number of distinct works in which both places appear), and it was this granularity that was selected as the appropriate level of analysis for the topology network.

Finding 4: Narrative topology diverges from geographic topology. Among the five test authors' works, the strongest co-occurring place pair is Leith and Princes

Street, with a co-occurrence weight of 28: they appear together in 28 distinct literary works. This weight is used throughout as the measure of narrative proximity – two places are narratively proximate if they co-occur frequently across works, regardless of physical distance. Geographically, however, the two are separated by approximately 2.5 kilometres, Leith a port district to the north and Princes Street the main commercial thoroughfare of the city centre, so their narrative proximity plainly does not reflect their spatial proximity. This is the clearest empirical illustration of the central claim behind Research Question 2.

Finding 5: Walter Scott's broad geographic range creates a display problem. Scott's works reference locations far outside the city centre (Forth, Haddington, Linlithgow), which, plotted against a common coordinate scale in Small Multiples, compress the city-centre data for all five authors into an unreadable band. The display range was fixed to the Edinburgh core (latitude 55.85–56.02, longitude -3.35 to -3.05), excluding out-of-range points from the map panel while retaining them in the sector-based analyses.

4.3 Implementation Technology

D3.js [4] was chosen for its fine-grained control over visual encoding, which the custom interactive designs in both directions require, and because it is web-native, requiring no installation or server infrastructure – important for sharing prototypes with the supervisor and study participants.

4.4 Spatial Classification: Neighbourhood Sectors

A key methodological decision in Direction 1 was how to define the spatial units (sectors) that would serve as comparable axes across authors. Three requirements guided this decision: the sectors must be few enough to serve as legible visual axes; they must be grounded in an authoritative external source so that the classification can be justified independently of the visualisation; and they must carry cultural and historical meaning relevant to literary analysis, in the sense that sector boundaries should not split apart areas the literary tradition treats as distinct (as with Old Town and New Town, discussed below), nor group together areas with unrelated literary associations.

The solution adopted is a two-level framework derived entirely from official City of Edinburgh Council data (Open Spatial Data Portal, Open Government Licence v3.0).

The primary layer is the **Neighbourhood Partnership Areas** (NP Areas), 12 legally established, administratively defined zones used by the Council for service planning. One NP Area, City Centre, was subdivided using a second official dataset, the **Natural Neighbourhoods** layer (resident-defined boundaries from a 2004 ward review, updated via 2014 public consultation), since it encompasses two zones historically and culturally distinct in the Edinburgh literary tradition: Old Town (the medieval city of the Royal Mile) and New Town (the Georgian planned city from the 1760s), jointly a UNESCO World Heritage Site but with substantially different literary characters – Old Town the setting for Scott's historical fiction and Stevenson/Hogg's Gothic sensibility, New Town the milieu of McCall Smith's contemporary fiction and Lockhart's biography. Conflating them into one City Centre sector would erase the corpus's most significant literary-geographic distinction. A third Natural Neighbourhood unit, Canongate, was also retained separately, as the eastern end of the Royal Mile with distinct historical significance.

The resulting framework comprises **14 sectors**: 11 unchanged NP Areas (Almond, Craigentinny/Duddingston, Forth, Inverleith, Leith, Liberton/Gilmerton, Pentlands, Portobello/Craigmillar, South Central, South West, and Western Edinburgh) plus Old Town, New Town, and Canongate derived from Natural Neighbourhoods. Place-name assignment proceeds by point-in-polygon intersection using the Shapely library.[1] Of the 2,135 place names, 1,480 (69%) fall precisely within a sector polygon; 423 (20%) lie in inter-polygon gaps within the Edinburgh bounding box and are assigned to the nearest sector; and 232 (11%) lie outside Edinburgh entirely and are excluded from sector-based analyses. Reuse of this Council data requires the following attribution under the Open Government Licence.[2]

4.5 Direction 1: Author Spatial Fingerprint

Three visualisation designs were developed and implemented for Direction 1.

4.5.1 Radar chart (primary design).

The radar chart represents each author as a polygon across the 14 sector axes, where the radius at each axis encodes the proportion of that author's total mentions that fall

[1] `https://shapely.readthedocs.io/`

[2] "Copyright City of Edinburgh Council, contains Ordnance Survey data © Crown copyright."

within the corresponding sector. The polygon shape is the author's spatial fingerprint. The chart was scaled to the maximum value in the dataset (0.436, McCall Smith's New Town proportion) rather than to 1.0, because fixing the axis to 100% leaves over 60% of the radial space empty and compresses the meaningful variation into a small central region.

The primary interaction is point-level, addressing the supervisor's feedback that the visualisation should provide "low-level information upon interaction": hovering any vertex reveals the sector percentage and its contributing places/books, and a click opens a detail panel listing the top place names and books for that sector and author, with a hovered place row surfacing a real example sentence from `api_sentence`. The legend supports filtering (click to hide/show) and isolation (double-click) by author.

The radar chart is implemented using D3's `lineRadial` function with `curveLinearClosed`, producing a closed polygon across the 14 angular positions ($2\pi/14$ radians per sector), with the radial scale mapping sector proportion to a radius between 0 and $R = 250$ pixels, capped at the observed data maximum (0.436) rather than 1.0.

Hover interaction uses invisible `circle` targets (radius 14px) centred on each vertex: a tooltip shows the sector name and percentage on hover, and a click opens a detail panel with the top eight places in that sector, ordered by mention count, plus a bar chart and the top five books.

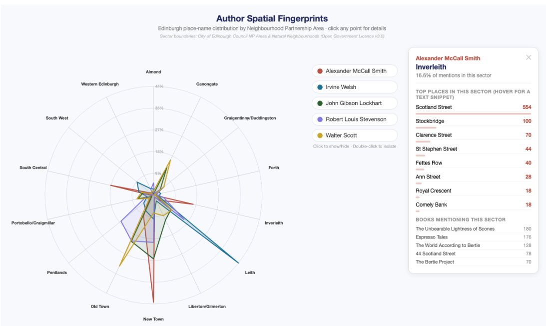

Figure 4.1: Clicking a vertex on the radar chart (here, Alexander McCall Smith's Inverleith point) enlarges the point and opens a detail panel showing the sector percentage, ranked place names, and the books mentioning that sector.

4.5.2 Bar-code fingerprint (secondary design).

The bar-code view represents each author as a horizontal row of vertical bars, one bar per place name, ordered from left to right by sector (north to south). The height of each bar encodes the proportion of that author's total mentions attributable to the corresponding place.

The y-axis scaling went through three iterations. A first logarithmic-scale prototype was dropped after supervisor feedback that it is difficult for non-specialists to interpret and that distribution shape mattered more than absolute magnitude here. A second, linear scale shared across all five authors ran into a different problem: Welsh's Leith alone accounts for 44.3% of his mentions, so on a shared axis every other author's distribution compressed into under a quarter of the chart height. The final design uses a per-author independent Y-axis, prioritising within-author shape over cross-author magnitude comparison; the trade-off is flagged by an interface warning, and a shared-axis toggle remains available for when cross-author comparison is wanted.

The bar-code view provides three ordering modes (by sector, by frequency, alphabetical) and two scale modes (percentage, absolute mention count), all switchable via controls at the top of the page. A text search field highlights the bars corresponding to a searched place name across all authors simultaneously, and a sector legend allows individual sectors to be hidden, narrowing the display to a subset of the city. Clicking any bar opens a detail panel listing the books that mention the selected place and a representative example sentence, mirroring the sentence-level detail added to the radar chart.

Each author's row is an SVG element with Y-axis domain `[0, maxVal]`, where `maxVal` is that author's own maximum proportion rather than the global maximum – the deliberate per-author scaling justified above.

Sector boundaries use two complementary encodings: a low-opacity background rectangle spans each sector's bars, and a narrow colour strip below identifies it, with text labels shown only where a sector is wide enough to hold them (threshold 40px). Switching ordering or scale mode re-renders the SVG elements directly rather than animating them, avoiding the complexity of D3 transition management across multiple groups.

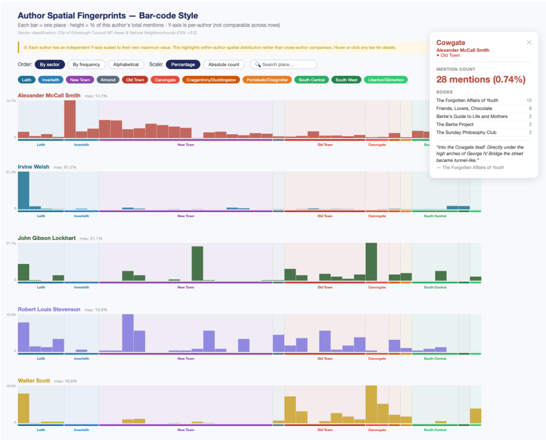

Figure 4.2: Clicking a bar (here, McCall Smith's Cowgate bar) opens a detail panel with the mention count, contributing books, and an example sentence.

4.5.3 Small multiples (tertiary design).

Five side-by-side Leaflet.js maps [10] (CartoDB Positron basemap [5]), bubble size encoding mention frequency (log by default, switchable to linear) and colour identifying the author. Clicking a bubble opens a popup with place name, mention count, sector, and an example sentence; a sector-legend filter hides bubbles across all five panels at once; an optional synchronised-zoom mode locks all panels to the same viewport; and a "compare two authors" mode swaps the five-panel grid for two larger, independently zoomable maps for closer inspection of a chosen pair.

The CartoDB Positron style was selected over standard OpenStreetMap tiles because its reduced visual complexity (lighter colours, fewer labels) allows the data bubbles to remain visually prominent. The coordinate range is fixed to the Edinburgh core area as described in Finding 5 above.

Each of the five panels is an independent Leaflet.js map instance (CartoDB Positron tiles), with place-name bubbles as Leaflet `CircleMarker` objects whose radius recomputes on scale-mode change; this replaced an earlier two-stage Leaflet/D3 overlay that required manual screen-coordinate projection and did not handle pan/zoom correctly. Synchronised zoom attaches a `moveend` listener to each map that calls `setView` on the others when the sync toggle is enabled, guarded by an `isSyncing` flag against cascading updates.

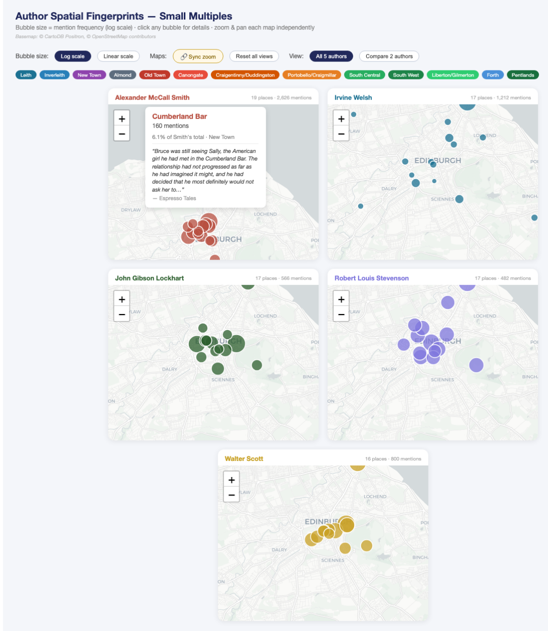

Figure 4.3: Clicking a bubble on McCall Smith's map (Cumberland Bar) opens a popup with the mention count, sector, percentage of the author's total, and an example sentence, over the CartoDB Positron basemap.

4.6 Direction 2: Narrative Topology Networks

Three visualisation designs were developed and implemented for Direction 2.

4.6.1 Force-directed network (primary design).

The force-directed network renders the document-level co-occurrence structure as a node-link diagram. Each node represents a place name, sized by total mention frequency across all authors; each edge represents a co-occurrence pair, with edge width encoding weight via $w' = (w/w_{\max})^2 \times 12$, chosen after a linear mapping left edges indistinguishable in width across the full weight range 1–28 and a square-root mapping still under-emphasised the strongest connections. The squared function produces a clear visual hierarchy while keeping weaker connections visible as thin lines.

Node positions are determined entirely by the force simulation (Fruchterman-Reingold [8] via D3's `forceSimulation` [4]), with no geographic component. A node's position in the layout reflects its pattern of co-occurrence connections, not its map location. This is a deliberate design choice: the diagram is meant to show narrative topology, and importing geographic position would conflate the two.

An interactive weight-threshold slider (range 1–28) allows the user to progressively reveal weaker connections, from the sparsest high-confidence view (threshold 28, showing only the single strongest pair) to the densest view (threshold 1, showing all 403 pairs). Nodes are draggable, and the diagram supports scroll-to-zoom and pan. Hovering over a node reveals a tooltip listing the five places most strongly co-occurring with it; clicking a node opens a detail panel with the same co-occurrence ranking, the books that mention the place, and an example sentence. A toggle switches node size between the absolute-frequency encoding described above and a percentage-of-total encoding, trading a compressed logarithmic scale for a more literal but more visually dominated-by-outliers area encoding.

The simulation uses four forces: `forceLink` (spring attraction, distance inversely proportional to weight), `forceManyBody` (uniform repulsion, strength -400), `forceCollide` (collision avoidance, node radius + 18px), and `forceCenter` (gentle centring). It re-runs from scratch on each weight-threshold change, since the active node/edge set changes and a smooth topology transition is not meaningful. Node labels sit at $y = \text{node.y} + r + 12\text{px}$, below each node; an earlier centroid placement with white fill was invisible against the equally dark node background.

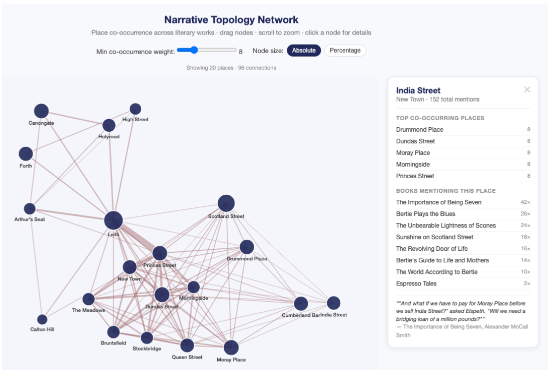

Figure 4.4: Clicking a node (India Street) opens a detail panel ranking its top co-occurring places and the books mentioning it, with an example sentence.

4.6.2 Linear connection diagram (secondary design).

The linear connection diagram arranges place names along a horizontal axis, ordered left to right by total mention frequency. Bézier arcs connect co-occurring pairs above the axis; arc height is proportional to the horizontal distance between the two connected nodes, and stroke width is proportional to co-occurrence weight. A weight-threshold slider (range 1–28) controls which pairs are shown, and the horizontal ordering can be switched between frequency, sector, and alphabetical. Clicking an arc opens a detail panel listing the books in which both connected places co-occur; hovering over a place reveals an example sentence. A search field highlights a named place across the diagram.

The linear rather than circular layout avoids implying the cyclical or symmetrical structure a chord diagram would (see below); the leftmost position (highest-frequency place) corresponds loosely to centrality in the corpus, so the left-to-right arrangement can be read as a rough ordering by narrative salience.

Each arc is drawn as a quadratic Bézier path: `M x1 y0 Q midX (y0 - h) x2 y0`, where $\texttt{midX} = (x1 + x2)/2$ and $\texttt{h} = |x2 - x1| \times 0.42$. The coefficient 0.42 was chosen to produce arcs that are visually distinct without extending beyond the SVG canvas for the longest connections. Stroke width is computed as $(w/w_{\max})^{1.5} \times 6$.

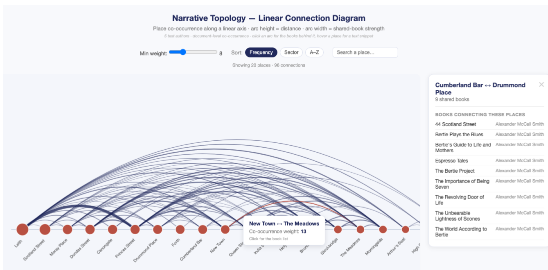

Figure 4.5: Clicking an arc (Cumberland Bar–Drummond Place) opens a detail panel listing the books connecting the two places, while hovering a place (New Town–The Meadows) shows the co-occurrence weight tooltip.

4.6.3 Metro-style map (illustrative design).

The metro-style map arranges place names as stations on a small number of named lines, in the visual language of Harry Beck's 1933 London Underground map, which replaced geographic accuracy with functional topology (which line to take, where to change) [2]. This project applies the same substitution to literary geography, with lines representing narrative corridors (clusters of places that co-occur strongly) rather than physical routes. Beck's map, however, still encoded a real network – the lines corresponded to physical tracks, only the geometry was distorted for legibility – a property an initial prototype of the literary metro map lacked: eight lines were defined manually along geographic and cultural axes, with station order following geographic rather than measured narrative sequence.

An audit of this prototype against the document-level co-occurrence data (Finding 3, Section 4.2) found that only 15 of 67 adjacent same-line station pairs (22%) shared at least one book in common: the map's line structure was a geographic argument wearing the visual grammar of a narrative one. The design was rebuilt around the following pipeline to close this gap.

(1) Community detection: modularity-based community detection using NetworkX's `greedy_modularity_communities` [9], which merges node clusters to maximise modularity, applied to the 57-place, 403-pair co-occurrence graph (weight ≥ 3), with communities over 18 stations recursively re-clustered. **(2) Station ordering**: a nearest-neighbour chain over real edge weights, so adjacency is backed by measured co-occurrence wherever possible. **(3) Interchanges**: a cross-line pair with weight ≥ 10 becomes a genuine interchange, a threshold chosen empirically for a legible 12 of 57 stations (a lower threshold gave 28, too dense to read as hubs). **(4) Line naming**: automatic, after a single author if they account for at least 55% of a line's book-mention volume (e.g. "Welsh's Edinburgh"), otherwise after its two highest-mention stations. **(5) Layout**: a co-occurrence-weighted spring embedding (NetworkX's `spring_layout`) gives each station a shared coordinate, so lines sharing an interchange meet at one point; positions are grid-snapped with collision avoidance. **(6) Octilinear correction and labels**: segments not aligned to Beck's 0°/45°/90° convention become a two-part "dog-leg"; labels are placed by a greedy eight-position search.

This pipeline reduces the map to five data-derived lines (rather than eight hand-drawn ones) and raises the proportion of adjacency backed by real co-occurrence from 22% to 88% (46 of 52 adjacent station pairs; per-line figures in Appendix B.5). The community structure also surfaced two findings invisible in the hand-drawn version, discussed further in Section 7: the word "Leith" clusters with the historical Old Town corridor associated with Scott and Lockhart rather than with the Welsh-dominated periphery line, and a small five-station line (Castle Street, Lasswade, the University of Edinburgh, the Royal Society of Edinburgh, and St Giles) emerged as a coherent biographical corridor, corresponding to the places Lockhart's *Life of Walter Scott* associates with Scott's own residences and professional life rather than with Scott's fiction.

The redesigned pipeline is implemented as two Python scripts – one deriving line structure (community detection, station ordering, interchange detection, layout) into a JSON file, the other rendering that JSON as the HTML map – so clustering parameters can be adjusted and the map regenerated without touching the rendering code. Because coordinates are shared across lines, two lines meeting at an interchange are drawn through the same physical point, producing genuine crossings rather than the separate parallel lanes of the initial prototype; interchange stations follow the London Underground convention (outer white ring, inner dark circle), and the greedy label-placement search reduced overlapping label pairs among the 57 stations from ten to two.

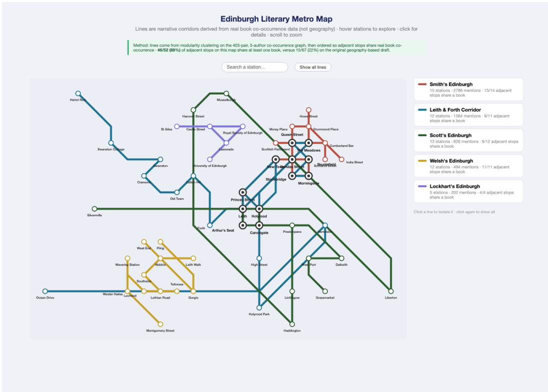

Figure 4.6: The metro-style map after the redesign: five lines derived from modularity clustering on the co-occurrence graph and named after their dominant author, with 88% of adjacent station pairs now backed by measured co-occurrence, against 22% for the initial hand-drawn prototype (Figure B.1, Appendix B.5).

4.7 Designs considered but not selected.

Two Direction-1 candidates were discussed but not implemented: a colour map silhouette (area shading over a simplified coastline), expected to produce visually similar heatmaps for all five authors since their mentions all concentrate in the city centre, giving limited discriminability; and a Direction-2 chord diagram, rejected before implementation after supervisor feedback that a circular arrangement implies cyclical structure, inappropriate for narratives with a definite beginning and end. Two further directions remain conceptual future work rather than implemented prototypes: a *Reader-plot timeline*, exploiting the `position_pct` field already built into `mention_order` to show place-name sequence within a single text (identified as future work, Chapter 8), and a *literary silences map* of places absent from the corpus, following Moretti [11], deferred because LitLong's coverage is not comprehensive enough to treat absence as meaningful evidence.

4.8 Design Summary

Table 4.1 summarises the six visualisation designs and their key properties.

Table 4.1: Summary of the six visualisation prototypes.

Direction	Design	Primary encoding	Key interaction
1	Radar chart	Polygon shape across 14 sector axes	Click vertex for sector detail panel; example sentence on hover
1	Bar-code	Bar height = mention share per place	Sector/frequency/alpha ordering; per-author Y-axis; search
1	Small multiples	Bubble size on real map	Sync zoom; log/linear toggle; sector filter; 2-author compare mode
2	Force-directed network	Node size = frequency; edge width = co-occurrence	Weight slider; drag nodes; click node for detail panel; absolute/percentage scale
2	Linear connection	Arc height = node distance; arc width = weight	Weight slider; sort by frequency/sector/alphabet; click arc for book list; search
2	Metro-style map	Line membership = community-detected narrative corridor	Click line to isolate; click station for detail panel; search

Chapter 5

Implementation

5.1 Technical Decisions and Constraints

Data embedding. All six D3 HTML files embed their data as JavaScript object literals – larger file sizes and a regeneration step on data change, an acceptable constraint for a research prototype evaluated on a fixed dataset.

Versioning and compatibility. Each HTML file's design history was tracked through numbered snapshots (e.g. `radar_v1.html`) during development, kept alongside working project files rather than in the submitted material. All files were tested in Safari and Chrome on macOS; D3.js v7 and Leaflet.js 1.9.4 load from CDN at runtime, requiring an internet connection.

Colour accessibility. The five fixed per-author colours were checked against simulated protanopia, deuteranopia, and tritanopia (the Brettel/Viénot transform) rather than assumed distinguishable. All ten pairs remain separable under all three; the closest, McCall Smith and Scott, is still above the confusion-risk threshold used here. The palette is reasonable but not maximally optimised, since colours were originally chosen for interface contrast rather than colour-vision-deficient legibility specifically.

5.2 Combined Interface

The combined interface[1] is a supplementary exploration tool, not part of the formal user study reported in Chapters 6–7: it puts all six designs behind one shared author selector

[1] Available live at `https://merrittw-1307.github.io/uoe-edinburgh-literary-geovis/data/processed/combined/d3/combined_interface.html`; source at `data/processed/combined/d3/combined_interface.html`.

(presets for 2/5/20/50/408 authors, free-text search, or a full custom checklist), reusing the client-side computation already built for the scale exploration (Appendix B.6). Five views (radar, bar-code, small multiples, network, linear) re-render live from the current selection; clicking an author in a fingerprint view highlights their places in a topology view via a shared `focusedAuthor` value, and a unified detail panel shows an example sentence and book list for whatever is hovered. The metro map is the one exception, since its lines depend on offline community detection (Section 4.6) rather than a per-keystroke computation, so it switches between five precomputed snapshots (2/5/20/50/408 authors) instead of recomputing live. At the 50-author preset the live co-occurrence graph reaches 12,833 edges, enough to stall D3's force simulation until rendering was capped to the 1,500 strongest edges – confirming, as a live interactivity constraint, the same density-versus-scale trade-off already reported for the standalone scale-exploration tool (Section 7.4).

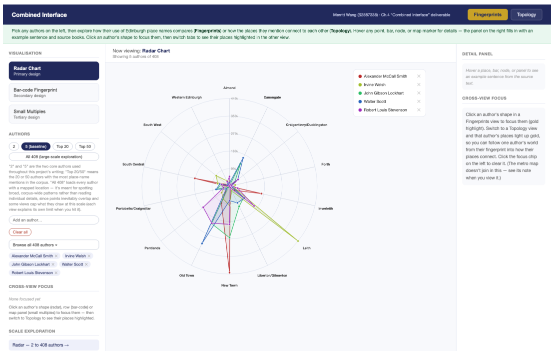

Figure 5.1: The combined interface's default view: a shared author selector (left) drives five re-rendering visualisation panels (centre) and a unified detail panel (right), shown here on the Radar Chart with the five test authors.

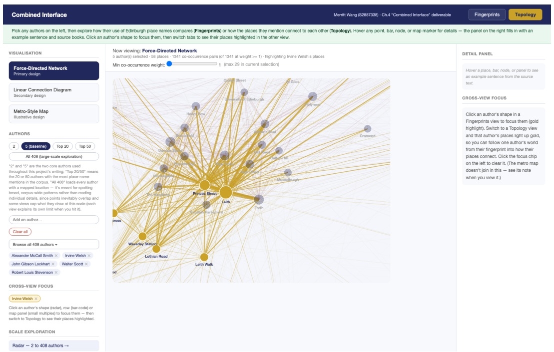

Figure 5.2: Cross-view focus: clicking Irvine Welsh's shape in the Fingerprints tab and switching to the Topology tab highlights his places in gold within the force-directed network, letting one author's spatial fingerprint be traced directly into the co-occurrence structure.

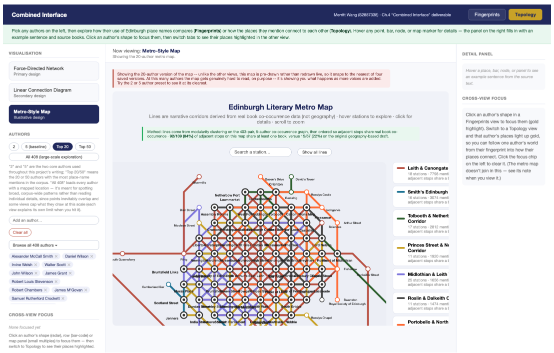

Figure 5.3: The metro map is the one view that does not re-render live: selecting the Top-20 preset snaps it to the nearest of five precomputed snapshots (here, the 20-author map) and surfaces a warning that legibility degrades by design as more authors are added.

Chapter 6

Evaluation

6.1 Study Design

The user study is designed to evaluate whether the six visualisation prototypes support the tasks implied by the two research questions: author identification (RQ1) and narrative-relationship identification (RQ2).

Participant groups. Two groups were recruited: domain experts (literary scholars, digital humanists, and researchers familiar with Edinburgh's literary geography) and general public, to test whether expert and non-expert users differ in which design they find most legible and in whether they spontaneously perceive the narrative/geographic divergence.

Format. The study was conducted online, unmoderated and self-paced, to maximise recruitment reach and to have participants interact with the visualisations in a real-world browser environment.

Introduction protocol. Participants saw each visualisation with only a brief description of the project's purpose, no explanation of the encoding or how to read it. This "cold" presentation tests whether the designs are self-explanatory, rather than participants' ability to apply an explanation they were given – a participant who needs the design explained is, in effect, reporting a usability problem. This differs from the labelled reference view in Task 1 below, which gives participants something to look at before being tested, without telling them what to look for.

Tasks. Two task types are used. Task 1 (fingerprint direction) tests the RQ1 success criterion from Chapter 1 – whether the fingerprint shapes are distinct enough to be learned and recognised, not whether a participant can name an author from memory of their books – via a study-then-identify structure. Participants first see a labelled reference view (all five authors' fingerprints together, the canonical prototype from Chapter 4) and spend a few minutes noticing how the shapes differ, unprompted. They then see a single-author blind variant with the legend and detail panel removed (which would otherwise expose book titles) and rendered in a neutral colour rather than the author's own, guarding against answering from a remembered colour swatch rather than the shape itself, and are asked which author it represents. To guard against a participant matching the outline from memory without engaging with what it communicates, the task opens with an objective, answer-keyed question – which sector or place the shape indicates the author writes about most – answerable only from the blind stimulus itself; the accompanying open-text question on what feature led to that answer is treated as equally important, since it captures reasoning even when the forced-choice answer is wrong. A different author is used per design to avoid anchoring across the three fingerprint tasks. Because a single blind identification only yields a one-in-five correct/incorrect judgement, Task 1 adds a third step: matching all five unlabelled, shuffled shapes to authors, producing a full 5×5 confusion matrix per design plus a self-rated confusability judgement. Both task directions also guard against ceiling/floor effects on their primary measures: Task 1 adds a runner-up guess and a forced choice between the single most confusable author pair per design (identified in advance by cosine similarity on the data); Task 2 adds ratings of three place pairs of known strong/medium/weak co-occurrence weight and a forced choice between a real data-derived cluster and two constructed distractors (full wording in Appendix A.3). Task 2 (topology direction) shows a topology visualisation and asks participants to identify the two most strongly connected places and judge whether they are geographically close or far; since all three topology designs already show every place labelled in one aggregated structure, no reference-then-blind structure is needed, and the secondary question tests whether the design spontaneously communicates the narrative/geographic divergence finding. After completing all tasks, participants vote for the most intuitive design in each direction.

Ethics. The study has received ethics approval from the Informatics Research Ethics Committee (application number 446635). Participants complete an online consent form before beginning the study. The participant information sheet and consent form are included in Appendix A.

6.2 Participants

Sixteen responses were collected; three were excluded on data-quality grounds (Section 6.3), leaving $n = 13$. Participants were recruited through the researcher's personal and academic network within the UK, per the eligibility criteria in the approved ethics application, not restricted to the University of Edinburgh.

Grouping into "domain expert" and "general public" was based on self-described background (Q-B3) rather than assigned at recruitment (Section 6.1): four participants reporting a background in literary studies, digital humanities, or visualisation/HCI/computer science formed the expert group, the remaining nine the general-public group – thinner on the expert side than the five-to-eight target, so treated throughout as exploratory rather than statistically powered.

A striking feature of the sample, expert group included, is near-uniform unfamiliarity with the five specific authors: nine of thirteen had read none of them, and eight rated themselves "not at all familiar" with Edinburgh's literary history, including the one literary-studies participant. This is not a recruitment weakness but a confirmation that the identification tasks (Section 6.4) test whether the encoding itself is learnable, not prior author knowledge. General visualisation-reading familiarity was more evenly spread ("moderately familiar" most common); four participants had lived in Edinburgh, one had visited, eight never had.

6.3 Study Procedure

The study was distributed as an anonymous Qualtrics link, completed online, unmoderated, and self-paced. Median completion time among the retained responses was 46 minutes, in line with the estimated 50–65; the wider recorded range reflects Qualtrics' "finish later" option letting some participants leave the survey open across more than one sitting, inflating elapsed but not active time.

Three of sixteen raw responses were excluded: one completed in under ten minutes with near-empty free text (a single full stop in place of an explanation); one gave single-letter, non-committal answers ("a", "b", "no") on every free-text field, including the anti-memorisation questions (F0, T1b) that specifically ask for justification; one completed in six minutes with free text unrelated to the questions asked. All three were judged not to reflect genuine engagement and excluded before analysis. A small number of retained free-text answers, originally given in Chinese by UK-resident participants, are translated by the researcher and reproduced alongside the original wording rather than replacing it.

6.4 Results

Fingerprint identification and matching (RQ1). Results below are reported per design, out of $n = 13$, with the single-author identity used for the blind stimulus given in parentheses.

Radar chart (Walter Scott). The encoding-legibility check (F0) was answered correctly by 10/13 participants (77%), the highest of the three designs. Single-shape identification (F1) was correct for 6/13 (46%), rising to 8/13 (62%) once a correct second guess (F1b) is credited. On the five-way matching task (M1), 45 of 65 shape-to-author assignments were correct (69%, against a 20% chance baseline), the highest matching accuracy of the three designs; the most common confusions were between Robert Louis Stevenson and Irvine Welsh, and between John Gibson Lockhart and Robert Louis Stevenson. On the targeted hardest-pair check (P1, Lockhart vs Scott), 11/13 (85%) correctly distinguished the two.

Bar-code (John Gibson Lockhart). F0 was correct for 6/13 (46%). F1 was correct for 6/13 (46%), rising to 10/13 (77%) on the Top-2 measure, the largest Top-1-to-Top-2 gain of the three designs, consistent with participants narrowing the shape to a plausible pair without landing on the exact author. M1 matching accuracy was 31/65 (48%); the most frequent confusions were between Robert Louis Stevenson and John Gibson Lockhart, and between Irvine Welsh and Walter Scott. P1 (Scott vs Welsh) was correct for 10/13 (77%).

Small multiples (Robert Louis Stevenson). This design produced the weakest results on every fingerprint measure: F0 correct for 5/13 (38%), F1 for 3/13 (23%), Top-2 for 5/13 (38%), and M1 matching accuracy 32/65 (49%), with Walter Scott the author most often confused for others. P1 (Lockhart vs Stevenson) was nonetheless correct for 10/13 (77%), comparable to the other two designs, suggesting that even where overall identification is weak, the single most-confusable pair is not disproportionately harder here than for the stronger designs.

Self-explanatoriness (F4, "would you have understood this without explanation") was mixed for all three designs, with no design rated "yes, easily" by more than a third of participants. In the post-task ranking, the three fingerprint designs were close to an even three-way split (small multiples 5 votes, radar chart 4, bar-code 4).

Narrative topology (RQ2). All three topology designs share the same underlying co-occurrence data, so the same ground-truth pair (Leith and Princes Street) and design-specific place pairs/clusters documented in Appendix A apply throughout.

Force-directed network. T1 (identify the strongest pair) was correct for 9/13 (69%). Ratings on the three fixed strength-check pairs (T-Strength) correlated strongly with the pairs' real co-occurrence weights ($r = 0.64$, $p < 0.001$, $n = 39$ ratings), the strongest correlation of the three designs. T-ClusterVerify was correct for 9/13 (69%), matching T1's accuracy.

Linear connection diagram. T1 was also correct for 9/13 (69%), identical to the network. The strength-rating correlation was weaker and not conventionally significant ($r = 0.30$, $p = 0.067$, $n = 39$). T-ClusterVerify accuracy was markedly lower than the other two designs at 3/13 (23%); the large majority (8/13) instead selected a distractor grouping (New Town, Leith Walk, Holyrood, Waverley Station) built by deliberately mixing places from different real communities, suggesting the linear diagram's arc-based encoding does not communicate cluster membership as clearly as the other two designs, despite matching them on the single strongest-pair task.

Metro-style map. T1 accuracy was substantially lower than the other two designs at 2/13 (15%), even though the same underlying pair is, by construction, the single strongest edge in every design. The strength-rating correlation was weak but statistically significant ($r = 0.34$, $p = 0.035$, $n = 39$). T-ClusterVerify accuracy was 7/13 (54%), between the other two designs. Despite this being the weakest design on two of the three objective measures, it was the clear winner of the post-task ranking (7/13 votes, against 5 for the network and 1 for the linear diagram), and of the final "most memorable visualisation" question (Q-R5) among those who named a specific design.

6.5 Discussion of Results

The clearest single result is that the radar chart outperforms the other two fingerprint designs on every measure: F0 comprehension, single-identification and Top-2 accuracy, five-way matching, and hardest-pair discrimination. This matches the qualitative pattern in F3 answers, where radar-chart responses often named a specific sector ("a very large spike in Leith area", "the maximum is old town"), while bar-code and small-multiples answers more often described a feature without naming it ("peak value"). The polygon's asymmetry appears to give participants a concrete verbal handle the other two encodings did not as reliably provide; since F0 accuracy tracks F1 accuracy design-for-design, this supports F0 as a genuine engagement check rather than a proxy for shape memorability alone.

Small multiples' weak performance across F0, F1, and M1 is notable given it is the most visually distinctive design at the reference-viewing stage (five separate maps): that distinctiveness evidently did not transfer to the blind, isolated shape. All three designs nonetheless cleared the P1 hardest-pair check well above chance (77–85%), and the gap between M1 and P1 for small multiples specifically (49% vs 77%) indicates its difficulty lies more in distinguishing all five patterns at once than in any single pairwise comparison.

For topology, the network and linear diagram matched on the single strongest-pair task (69% each) but diverged sharply on cluster recognition (network 69%, linear 23%, barely above the 33% random baseline). Read with the network's stronger strength-rating correlation ($r = 0.64$ against $r = 0.30$), this points to one difference: encoding strength as edge thickness and spatial proximity communicates both a strongest pair and a broader cluster more reliably than arc height along a single axis, even though the two are statistically indistinguishable on the narrower single-pair task.

The metro map is the most interesting tension in the results: worst of the three on the strongest-pair task (15%, against 69% for the others) despite encoding the same weight in its line structure, yet decisively the most popular in the post-task ranking and most-named "most memorable" design. Participants' own explanations ("similar to real transit maps I am familiar with", "its like a map that we see everyday") suggest its borrowed visual language reads fluently as a familiar transit map rather than as an instrument for locating one specific edge – confident navigation, not task accuracy. This is an unintended illustration of the same trade-off Beck's Underground map itself makes (Section 4.6): a comfortable, generalisable mental model of the whole network over precise retrieval of any single edge weight.

Cross-tabulating by expert/general-public group carries the caveat, restated because it matters, that four experts is too small a group for a robust comparison (a single answer shifts the expert percentage by 25 points). Descriptively, expert free-text answers were not obviously more accurate, but more often referenced the underlying literary-geographic content directly: one expert's synthesis answer, "I was surprised that some places that are geographically quite separate still seem closely connected in the authors' writing. It made me realise that literary connections between places do not necessarily follow their physical locations," restates the dissertation's own RQ2 argument unprompted, and another's, "Yes, Walter Scott in Leith I did not realise this is where his work was set," is a direct instance of the visualisation surfacing an unknown literary fact. Both are treated as indicative rather than representative given the sample size, but are the kind of qualitative evidence the deep-insight questions (T3, T4, Cross-design synthesis) were designed to surface.

Chapter 7

Discussion

7.1 Design Improvements after User Study

The results in Section 6.4 point to concrete, design-specific changes rather than one across-the-board fix, since each design failed, where it failed, differently.

For small multiples, the weakest performer on every objective measure, the gap between strong reference-view distinctiveness and weak blind-identification accuracy suggests the problem is the loss of side-by-side comparison rather than legibility in the abstract: a faint, unlabelled silhouette of the other four authors' spatial envelopes in the blind view is worth testing, preserving comparison without reintroducing identity. For bar-code, the large Top-1-to-Top-2 gain (46% to 77%) suggests participants extract real signal without a reliable way to commit to one answer; defaulting to frequency rather than sector ordering, already several participants' stated F3 strategy, may make the largest bar easier to commit to.

For topology, the metro map's weak strongest-pair accuracy despite strong subjective preference is the clearest case for a targeted change: since its transit-map familiarity is a genuine strength worth keeping, a lightweight strength cue within that visual language (line thickness, or station size proportional to interchange weight) is more promising than abandoning the metaphor. The linear diagram's poor cluster recognition (23%, barely above the 33% chance baseline) despite matching single-pair accuracy suggests its single-axis layout communicates one connection well but not multi-place groupings; arc-bundling or community-membership colour-coding would give it an explicit cluster cue it currently lacks.

More generally, F0's correlation with F1 accuracy across all three fingerprint designs suggests future iterations could use F0-style objective legibility checks earlier in the design process, catching a distinctive-looking but non-communicative design before a full user study is built around it.

7.2 Narrative Topology vs Geographic Topology

The most striking finding from the data analysis is the co-occurrence of Leith and Princes Street across 28 distinct literary works, the strongest pair in the entire corpus, despite sitting roughly 2.5 kilometres apart: Leith a port district historically tied to trade and working-class life, Princes Street the commercial artery of the bourgeois Georgian New Town. On a geographic map they are simply two points in the same city; in the narrative topology visualisation they are the closest pair in the network. One plausible reading is that the journey between them recurs as a structuring device in Edinburgh fiction because it crosses a social as well as a spatial boundary – consistent with Irvine Welsh's *Trainspotting* and Scott's historical novels – though this is offered as a literary reading of the co-occurrence pattern, not a claim tested directly by the data. A geographic map cannot show this; the narrative topology visualisation shows these two places as *narratively adjacent* regardless of spatial distance, speaking directly to Westphal's geocriticism, which argues literary space is relational and constructed rather than a reflection of physical geography [19]: the places most strongly connected in Edinburgh's literary imagination are not the places nearest each other on a map.

The metro-map redesign (Section 4.6) offers a related but methodologically distinct observation: an internal-consistency check rather than an independent test of the claim above, since the redesigned lines are themselves derived from the co-occurrence graph and are then checked against that same graph. Read this way, the result is still informative – only 22% of the hand-drawn map's same-line adjacencies were backed by real co-occurrence, against 88% once lines were derived from the graph itself – but as evidence that a co-occurrence-optimised grouping is more internally consistent with the underlying data than a geography-optimised one, consistent with Finding 4, rather than as a second independent demonstration that narrative topology diverges from geography (the Leith/Princes Street finding above, which compares co-occurrence against an unrelated measure, geographic distance, is the independent evidence for that claim). The redesign also surfaced two findings the geographic version could never have produced. Leith, as a place name across all five authors combined, groups with the historical Old Town corridor of Scott and Lockhart rather than with the contemporary, working-class Leith of Welsh's novels – a finding about how the name functions across a corpus rather than about any one author's usage, visible only once co-occurrence rather than individual usage is what groups places. Separately, a five-station corridor (Castle Street, Lasswade, the University of Edinburgh, the Royal Society of Edinburgh, and St Giles) emerged as a coherent cluster corresponding to Lockhart's biographical rather than fictional geography of Scott – places he lived and worked, per *Life of Walter Scott*, not places his novels are set – distinguishing this biographical register from the historical-fiction register of the neighbouring Old Town corridor without being told to.

7.3 Author Spatial Languages

The near-total absence of shared place names across the five test authors (zero shared by three or more, only 18 by any two) can be stated as a simple number but deserves interpretation: each author has constructed an Edinburgh that is almost entirely their own – McCall Smith's bounded by the Scotland Street/Dundas Street/Drummond Place triangle in the New Town, Welsh's by Leith, Scott's by the Canongate and Old Town closes, Stevenson's extending to the Pentlands (his childhood at Swanston Cottage) and the Firth of Forth (his father's lighthouse engineering). The radar chart and bar-code fingerprint make these private geographies comparable at a glance – a user who knows nothing of these authors' biographies can see, from polygon shape alone, that McCall Smith and Welsh occupy different spatial territories within the same city, encoding in geometry an argument that would otherwise require paragraphs of prose.

This vocabulary separation also explains a non-obvious pattern from scale-testing the network graph against the full corpus (`network_scale_explore.html`): co-occurrence density does not fall smoothly as author count grows, but is high at both the very small end (two authors, or two-to-three books) and low at the very large end (408 authors), for two related reasons. At the small end, document-level co-occurrence approaches its degenerate case, since almost any two places in a handful of books will co-occur at least once; three books (two by McCall Smith, one by Welsh) produced 152 places and 5,889 pairs at weight ≥ 1, collapsing to 33 places and 424 pairs at weight ≥ 2, almost all connecting McCall Smith's two books to each other rather than to Welsh's – because his fictional Scotland Street recurs across his own works while Welsh's Leith does not recur in either, a book-level instance of the same vocabulary separation. At the large end, the opposite problem appears: adding authors adds nodes far faster than genuine cross-author edges, so the 408-author graph is large (533 places, 23,380 pairs) but sparse (16.5% of possible pairs, against 79.6% for two authors). The five-author baseline used throughout sits between these failure modes, with enough authors for a cross-author edge to read as a meaningful, comparatively rare signal.

7.4 Limitations and Open Questions

Corpus size confound. McCall Smith contributed 17 works to the corpus, Stevenson 16, Scott 12; Welsh only 8, Lockhart 6. Because sector distributions are normalised as proportions of each author's total mentions, a large and a small corpus with identical spatial habits would produce identical fingerprints – worth being upfront about. That normalisation, though, means results are best read as measures of *spatial attention* rather than absolute output, and the consistency of spatial patterns across works within each author's corpus suggests these are genuine authorial preferences rather than an artefact of corpus size.

Five-author prototype, tested at scale. The current system uses five test authors, selected for contrast and representativeness; rather than leave open whether this scales to the full 408-author corpus, all six designs were tested directly using scale-exploration tools kept separate from the six canonical prototypes (Appendix B.6). The six designs do not share one scaling bottleneck: radar and bar-code (Direction 1) both lose the "compare everything at a glance" property, but radar becomes visually unreadable while bar-code grows navigationally costly instead (roughly 50 screens of scrolling); network and linear (Direction 2) both grow along one spatial dimension, but network degrades gracefully while linear's canvas simply outgrows the viewport; small multiples alone has a genuine computational cost, degrading 3.6$\times$ from 50 to 408 authors; and metro's failure sits in the clustering algorithm itself, with adjacency-backed accuracy declining from 96% at 2 authors to 72% at 50. Full per-design evidence is in Table B.6.

Metro-map layout is heuristic, not optimal. The metro-map redesign (Section 4.6) replaces manual line assignment with community detection and manual station placement with a spring embedding corrected for octilinearity; neither solves the underlying schematic-layout problem exactly, since Nöllenburg and Wolff's result [13] shows an exact solution is computationally hard in general. Two residual limitations follow: the line structure is sensitive to the community-detection threshold and the five-author, top-15-places-per-author filter used to build the graph (re-running the pipeline at larger N, Appendix B.6, confirms this, so the current five lines are a favourable point on the scaling curve rather than a fixed taxonomy), and the greedy label-placement search leaves two of 57 station labels overlapping in the densest interchange cluster, which a fully general solver or manual adjustment could resolve.

Source text sampling. All six prototypes surface example sentences from `api_sentence` (Section 5), so readers see the literary sentence a place appears in rather than only an aggregate count. The remaining limitation is that examples are sampled (fixed random seed, for reproducibility) rather than curated, so a place with many mentions shows an arbitrary rather than necessarily representative subset – a curated or relevance-ranked selection is left for the combined interface.

Reader-sketch and narrative time. A further direction is visualising the reading experience itself: the sequence in which places are encountered as the reader moves through a text, rather than their aggregate frequency or co-occurrence. The `position_pct` field in `mention_order` provides the necessary data for this, and Reader-plot timeline visualisations were prototyped but not included in the user study. This remains an open direction.

User study sample size. The evaluation draws on $n = 13$ participants, below the 13–20 planned during recruitment and well below what supports statistically powered comparisons; percentages in Chapter 6 should be read as directional, with a single participant worth roughly 8 percentage points on any measure. The expert subgroup ($n = 4$) is thinner still, so expert/general-public comparisons (Section 6.4) are indicative rather than conclusive. Recruitment was convenience-based, through the researcher's own network rather than a public call, likely skewing the sample toward participants favourably disposed to engage carefully – a plausible contributor to the generally low self-reported confusion rates (Q-R3) – so a larger, more broadly recruited sample would be needed before drawing firm conclusions about generalisability. The exclusion of three low-quality responses (Section 6.3) used researcher judgement against criteria not formally pre-registered before data collection, a standard limitation of a small, time-constrained study, stated here for transparency.

Chapter 8

Conclusion

This dissertation has argued that place names in literary text are data about narrative structure rather than merely geographic location, and has shown that this argument can be operationalised as a set of interactive visualisations grounded in real corpus data. The LitLong Edinburgh database, 620 literary works spanning three centuries, 2,135 place names, 50,248 mention records, provided the empirical foundation, and six visualisation prototypes were developed across two research directions, implemented in D3.js, and evaluated through a user study.

RQ2 — does narrative co-occurrence reveal a topology distinct from geography? Yes. The strongest evidence is the Leith/Princes Street pair itself: 28 shared works despite roughly 2.5km of geographic separation, one of only 12 station-pairs across the redesigned metro map that behave as a genuine narrative interchange. The metro-map redesign (Section 4.6) turned this from a single anecdote into a systematic internal-consistency diagnostic, deriving line structure from measured co-occurrence rather than hand-drawn geography and raising the proportion of adjacent stations sharing a real book from 22% to 88% – not independent proof of the divergence above, since the lines are themselves built from the co-occurrence data being checked (Chapter 7), but a systematic confirmation that a co-occurrence-optimised structure is more internally consistent with the data than a geography-optimised one, while surfacing two further instances of the same divergence geography could not have produced: "Leith" clustering with Scott and Lockhart's historical corridor rather than Welsh's, and a biographical corridor (Castle Street, Lasswade, the University of Edinburgh) distinct from the historical-fiction corridor next to it. The scale-exploration work (Appendix B.6) adds a boundary condition rather than a contradiction here: the divergence is clearest at the five-author scale used throughout, since the same near-total non-overlap of authorial vocabularies (Finding 1) that drives it also thins the co-occurrence signal at very large author counts and degenerates it at very small ones – most legible at a middle scale, not a universal property independent of N.

RQ1 — does an author's spatial distribution form a visually distinctive, learnable fingerprint? Partially yes. Across the user study ($n = 13$), the radar chart gave the strongest evidence: after studying all five authors' fingerprints together, participants matched all five unlabelled shapes back to their authors at 69% accuracy (20% chance baseline), and distinguished the single most similar pair 85% of the time. Bar-code and small-multiples showed the same direction of effect more weakly (48% and 49% matching accuracy), and single-shape identification in isolation (23–46%) was consistently harder than matching all five side by side – these fingerprints are more reliably told apart from each other than recognised from memory alone. The evidence supports RQ1 for at least the radar chart, without supporting a stronger claim that fingerprinting works equally well regardless of encoding; which choices explain the radar chart's advantage, and whether it holds at a larger sample, is discussed in Chapter 7.

The project's contributions include a spatial classification of Edinburgh place names derived from official Council boundary data; the finding that each of the five selected authors constructs an almost entirely private imaginary geography of the city; a method for extracting narrative reading order from `start_word` as a monotonic index; and a suite of interactive prototypes making authorial spatial patterns and narrative co-occurrence structure visible in forms not previously available.

Future work: a faster, client-side approximate clustering method could free the metro map from its precomputed author-count snapshots (Section 5.2), and a single detail panel shared consistently across all six views would be a natural consolidation. Beyond the interface, the `api_posmention` table's part-of-speech annotations would support narrative weight analysis (dialogue vs. narration vs. description, and whether this varies by author or genre), and "literary silences," Edinburgh places absent from any literary work, would complement the current analysis.

Bibliography

[1] Isabelle Anderson, David Beavan, Beatrice Costa, Aaron Quigley, Tim Harris, and Thomas Corns. Palimpsest: Literary Edinburgh and the Use of the Geo-annotation in Literature. In *Digital Humanities 2016: Conference Abstracts*, Krakow, Poland, 2016.

[2] Harry Beck. London underground map, 1933. Transport for London.

[3] David J. Bodenhamer, John Corrigan, and Trevor M. Harris, editors. *The Spatial Humanities: GIS and the Future of Humanities Scholarship*. Indiana University Press, Bloomington, IN, 2010.

[4] Michael Bostock, Vadim Ogievetsky, and Jeffrey Heer. D3: Data-driven documents. *IEEE Transactions on Visualization and Computer Graphics*, 17(12):2301–2309, 2011.

[5] CARTO. Positron basemap, 2024. `https://carto.com/basemaps`.

[6] David Cooper and Ian N. Gregory. Mapping the English Lake District: A literary GIS. *Transactions of the Institute of British Geographers*, 36(1):89–108, 2011.

[7] Johanna Drucker. Humanities approaches to graphical display. *Digital Humanities Quarterly*, 5(1), 2011.

[8] Thomas M. J. Fruchterman and Edward M. Reingold. Graph drawing by force-directed placement. *Software: Practice and Experience*, 21(11):1129–1164, 1991.

[9] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th Python in Science Conference (SciPy2008)*, pages 11–15, 2008.

[10] Leaflet. Leaflet — an open-source javascript library for interactive maps, 2024. `https://leafletjs.com/`.

[11] Franco Moretti. *Graphs, Maps, Trees: Abstract Models for a Literary History*. Verso, London, 2005.

[12] Tamara Munzner. *Visualization Analysis and Design*. A K Peters/CRC Press, Boca Raton, FL, 2014.

[13] Martin Nöllenburg and Alexander Wolff. Drawing and labeling high-quality metro maps by mixed-integer programming. *IEEE Transactions on Visualization and Computer Graphics*, 17(5):626–641, 2011.

[14] Barbara Piatti, Hans Rudolf Bär, Anne-Kathrin Reuschel, Lorenz Hurni, and William Cartwright. Mapping literature: Towards a geography of fiction. In *Cartography and Art*, pages 1–16. Springer, Berlin, Heidelberg, 2009.

[15] Ben Shneiderman. The eyes have it: A task by data type taxonomy for information visualizations. In *Proceedings of the IEEE Symposium on Visual Languages*, pages 336–343, 1996.

[16] Jan-Erik Stange and Marian Dörk. Unfolding the city: Visualizing urban narratives and stories. In *Proceedings of the 7th International Conference on Information Visualization Theory and Applications*, pages 91–98, 2015.

[17] Robert T. Tally. *Spatiality*. Routledge, London, 2013.

[18] Edward R. Tufte. *The Visual Display of Quantitative Information*. Graphics Press, Cheshire, CT, 1983.

[19] Bertrand Westphal. *Geocriticism: Real and Fictional Spaces*. Palgrave Macmillan, New York, 2011.

Appendix A

Study Material

A.1 Participant Information Sheet

Participant Information Sheet

Project title:	Studying Experiences with Data Representations of a Collection of Edinburgh-related Literature
Principal investigator:	Uta Hinrichs
Researcher collecting data:	Merritt Wang
Funder (if applicable):	N/A

This study was certified according to the Informatics Research Ethics Process, reference number 446635. Please take time to read the following information carefully. You should keep this page for your records.

Who are the researchers?

This research conducted by Merritt Wang as part of their MSc project at the School of Informatics, University of Edinburgh. The MSc project is supervised by Dr. Uta Hinrichs, Reader in Data Visualization at the School of Informatics.

What is the purpose of the study?

This MSc project focuses utilizing data representations (e.g., data visualizations or physicalizations to make a collection of Edinburgh-related literature explorable from different perspectives. As part of this study we would like to explore people's interest in the data and their experiences of bespoke data representation.

Why have I been asked to take part?

You have been asked to take part in this study as you live in Edinburgh, have indicated an interest in Edinburgh-related literature, and/or have expressed an interest in this project. Your participation in this study will help us design and refine relevant data representations to make this collection of Edinburgh-related literature explorable by a wide range of audiences.

Do I have to take part?

No – participation in this study is entirely up to you. You can withdraw from the study at any time, up until 7 days after your study session without giving a reason. After this point, personal data will be deleted and anonymised data will be combined such that it is impossible to remove individual information from the analysis. Your rights will not be affected. If you wish to withdraw, contact the PI – Dr. Uta Hinrichs (uhinrich@ed.ac.uk). We will keep copies of your original consent, and of your withdrawal request.

What will happen if I decide to take part?

You will first be asked to complete a short background questionnaire about your interest in or experience with Edinburgh-related literature and/or data visualisation (approx. 5 min.). You will then be introduced to several data visualisations representing Edinburgh-related literature and asked to complete a small number of tasks based on them (for example, identifying which author a visualisation belongs to, or judging the relationship between places), followed by a short reflection questionnaire about your experience. Your task responses and questionnaire answers will be recorded in writing; no audio or video recording will take place. The full session should take approx. 35-50 min. to complete.

Are there any risks associated with taking part?

There are no significant risks associated with participation.

Are there any benefits associated with taking part?

There are no direct benefits for participating in these study activities, other than learning about Edinburgh-related literature and getting the opportunity of informing ways of exploring it.

Refreshments will be provided for in-person study activities.

What will happen to the results of this study?

The results of this study may be summarised in published articles, reports and presentations. Quotes or key findings will be anonymized: We will remove any information that could, in our assessment, allow anyone to identify you. With your consent, information can also be used for future research. Your data may be archived for a maximum of 4 years. All potentially identifiable data will be deleted within this timeframe if it has not already been deleted as part of anonymization.

Data protection and confidentiality.

Your data will be processed in accordance with Data Protection Law. All information collected about you will be kept strictly confidential. Your data will be referred to by a unique participant number rather than by name. Your data will only be viewed by the researcher/research team named on this PIS.

All electronic data will be stored on a password-protected encrypted computer, on the School of Informatics' secure file servers, or on the University's secure encrypted cloud storage services (DataShare, ownCloud, or Sharepoint) and all paper records will be stored in a locked filing cabinet in the PI's office. Your consent information will be kept separately from your responses in order to minimise risk.

What are my data protection rights?

The University of Edinburgh is a Data Controller for the information you provide. You have the right to access information held about you. Your right of access can be exercised in accordance Data Protection Law. You also have other rights including rights of correction, erasure and objection. For more details, including the right to lodge a complaint with the Information Commissioner's Office, please visit www.ico.org.uk. Questions, comments and requests about your personal data can also be sent to the University Data Protection Officer at dpo@ed.ac.uk.

Who can I contact?

If you have any further questions about the study, please contact the lead researcher, Uta Hinrichs (uhinrich@ed.ac.uk).
If you wish to make a complaint about the study, please contact inf-ethics@inf.ed.ac.uk. When you contact us, please provide the study title and detail the nature of your complaint.

Alternative formats.

To request this document in an alternative format, such as large print or on coloured paper, please contact Uta Hinrichs (uhinrich@ed.ac.uk).

General information.

For general information about how we use your data, go to: Privacy notice | Edinburgh Research Office | Edinburgh Research Office

A.2 Consent Form

Participant number:______________________

Participant Consent Form

Project title:	Studying Experiences with Data Representations of a Collection of Edinburgh-related Literature
Principal investigator (PI):	Uta Hinrichs
Researcher:	Merritt Wang
PI contact details:	uhinrich@ed.ac.uk

By participating in the study you agree to take part in the study activities listed in the Participant Information Sheet.

- I have read and understood the Participant Information Sheet for the above study, that I have had the opportunity to ask questions, and that any questions I had were answered to my satisfaction.
- My participation is voluntary, and that I can withdraw at any time without giving a reason. Withdrawing will not affect any of my rights.
- I consent to my anonymised data being used in academic publications and presentations.
- I understand that my anonymised data will be stored for the duration outlined in the Participant Information Sheet.

Please tick yes or no for each of these statements.

1. I allow my data to be used in future ethically approved research.

Yes	**No**

2. I agree to take part in this study.

Yes	**No**

Name of person giving consent	Date dd/mm/yy	Signature
Name of person taking consent	Date dd/mm/yy	Signature

THE UNIVERSITY *of* EDINBURGH
informatics

A.3 Task Questions

The following instrument implements the two task types described in the Study Design section of Chapter 6, as distributed to participants via Qualtrics. Each participant saw all six visualisations; the order of the three fingerprint designs (and, separately, the three topology designs) was randomised per participant to control for ordering effects.

Task 1 — fingerprint identification and matching (shown once per fingerprint design: radar, bar-code, small multiples). Participants first see a reference view, the canonical all-five-authors version of the design in question, and are asked to spend a few minutes noticing how the five shapes differ before continuing; no hint is given as to what to look for. They then see a single-author variant of the same design, showing one author's shape with no legend or name anywhere, and answer:

1. Based on the chart shown above (not your memory of the reference view), which sector/place does this author's writing concentrate on the most? *(single choice among four options: the true answer plus the three next-highest sectors/places for that author by real value, so the distractors are genuinely close rather than arbitrary)*

2. You are now looking at one of the five fingerprints you just saw, with the name hidden. Which of the following five authors do you think it represents? *(Alexander McCall Smith / Irvine Welsh / John Gibson Lockhart / Walter Scott / Robert Louis Stevenson / I don't know)*

3. If your first guess turned out to be wrong, which of the five would be your second guess? *(same five authors, forced choice)*

4. How confident are you in your first answer? *(Not at all confident – Very confident, 5-point scale)*

5. What specific feature of the shape led you to that answer (for example, an unusually large spike in one area, or a shape that is spread evenly across many areas)?

6. If no one had told you anything about this visualisation, would you have understood how to read it just by looking? *(Yes, easily / Yes, with a little effort / No, I was confused)*

A single forced-choice identification (item 2) only ever yields a correct/incorrect judgement against a one-in-five baseline; the runner-up question (item 3) is added cheaply, from the same stimulus, to distinguish participants who narrowed the shape down to a plausible pair from those guessing entirely at random, since both would otherwise register identically as "incorrect." Item 1 addresses a different concern from items 2–6: even with the reference-view colour removed from the blind stimulus (Chapter 6), a participant could in principle answer item 2 correctly by matching the shape's overall silhouette against a memorised image from the reference view, without the shape's specific content – which sector or place it actually peaks toward – playing any role in that judgement, which would test recognition memory for an outline rather than comprehension of what the encoding communicates. Item 1 is answerable from the blind stimulus alone, with no memory of the reference view required at all, and is scored against a real answer key (the true highest-value sector or place for that author, against the next three highest as distractors) rather than the identification question in item 2, making it a direct, memorisation-resistant check of whether the encoding's actual content was read rather than its shape merely recognised.

Finally, participants see a third variant of the same design showing all five authors' shapes again, side by side, with no names and in a shuffled order distinct from the order used in the reference view, and answer:

7. Match each of the five unlabelled shapes to an author. *(matrix-table question: one row per shape, the same five authors as answer options in every row; a participant may assign the same author to more than one shape)*

8. Overall, having compared all five side by side, how easy or difficult do you think it would be to mix up any two of these five patterns? *(Very easy to confuse – Very easy to tell apart, 5-point scale)*

9. Look again at the two specific shapes identified (across all authors for this design) as most similar to one another by cosine similarity on their underlying value vectors. One is [Author A] and the other is [Author B]. Which is which? *(forced choice between the two orderings)*

The single-item identification question (item 2) yields only a correct/incorrect judgement against a one-in-five baseline; the matching question (item 7) instead yields a full 5×5 confusion matrix per design, indicating not just whether a design is distinguishable overall but which specific pairs of authors' fingerprints are most often mistaken for one another – a substantially richer test of the same RQ1 success criterion. The subjective rating (item 8) is compared against the matching question's objective accuracy as a check for over- or under-confidence in participants' own sense of how distinguishable the shapes are. Item 9 addresses a risk specific to item 7: if a design's five shapes happen to be very easy to tell apart overall, the confusion matrix can saturate near-perfect and say nothing about relative distinguishability between designs. Item 9 targets the single most confusable pair of authors for that design specifically – for the radar chart, John Gibson Lockhart and Walter Scott (cosine similarity 0.844 on their 14-sector value vectors, the highest of all ten author pairs); for the bar-code, Walter Scott and Irvine Welsh (the second-highest pair for that design, chosen instead of the top pair to avoid repeating the same two authors' names across two of the three fingerprint designs); for small multiples, John Gibson Lockhart and Robert Louis Stevenson (highest pair by cosine similarity on sector-aggregated mention counts) – guaranteeing a difficulty floor that a well-separated overall matrix cannot obscure, while the two-alternative format keeps chance performance at a known 50% rather than the one-in-five baseline of item 2. As with the single-author variant, the assignment of shapes to shuffled positions in the matching view is fixed rather than randomised per participant, since the matching stimulus is hosted outside the survey tool and there is no channel to record which shuffle a given participant saw; a different fixed shuffle is used for each of the three designs to prevent a participant from carrying a positional guess across designs.

Task 2 — narrative/geographic relationship (shown once per topology design: force-directed network, linear connection diagram, metro map). Participants are first invited to explore the visualisation freely (hover, click, zoom/scroll as the design allows) before answering, rather than being directed straight to the strongest pair; this is intended to surface spontaneous observations about the co-occurrence structure beyond the single forced-choice item, since a single near/far judgement on one pair alone would be too narrow a test of whether participants independently perceive the narrative-topology structure.

1. Looking at this visualisation, which two places appear to be the *most strongly connected*? *(free text, two place names)*

2. What about this diagram made you think these two places are the most connected (for example: the thickest line between them, how close together they are drawn, being on the same coloured line)? *(free text)*

3. Based on your own knowledge of Edinburgh (or your best guess if you are not

familiar with the city), are these two places geographically close together or far apart? *(Very close – Very far, 5-point scale / Not sure)*

4. For each of three specific place pairs, how strongly connected do they appear in this visualisation? *(matrix-table/rating-grid question, one row per pair, 5-point scale: Not connected at all – Extremely strongly connected)*

5. Beyond that pair, is there any *other* connection in this visualisation that surprises you — for example, two places that seem strongly linked despite seeming unrelated, or two places you would expect to be linked that do not appear connected? *(free text: place names + brief explanation, optional but encouraged)*

6. Do you notice any group of three or more places that seem to form a cluster of connections with each other? If so, list them and, if you have a guess, why you think they might be grouped together. *(free text, optional)*

7. Which of three candidate groupings of places looks like it forms the tightest cluster in this visualisation? *(single choice among three options, order randomised; one option is a genuine cluster, the other two are constructed by mixing places from different clusters)*

8. If no one had told you anything about this visualisation, would you have understood how to read it just by looking? *(Yes, easily / Yes, with a little effort / No, I was confused)*

Items 4 and 7 are added for the same reason as the fingerprint task's item 8 above: item 1 alone risks a ceiling effect if the strongest pair in a given dataset is visually unmistakable (here, Leith and Princes Street, at a co-occurrence weight of 28 against a next-highest weight of 18), and item 6 alone risks a floor effect if participants who notice nothing simply skip it, yielding no usable data either way. Item 4 asks participants to rate three place pairs deliberately chosen to span strong, medium, and weak ground-truth co-occurrence weight, rather than search for a pair themselves, producing interval-scaled data that can be correlated against the real weights regardless of whether item 1 saturates. Item 7 gives a forced choice between one real cluster and two distractors built by mixing places from different clusters, providing a scoreable recognition-based measure to sit alongside item 6's open-ended production task; agreement between what a participant names unprompted in item 6 and what they select in item 7 is treated as corroborating evidence in the analysis.

Item 2 addresses a separate risk specific to this task direction: the force-directed network, linear diagram, and metro map are three renderings of the *same* underlying co-occurrence weights, so the ground-truth answer to item 1 (and, before this revision, to items 4 and 7) is identical across all three designs. A participant who solves item 1 correctly on the first topology design they see – whether through genuine reading of that design or simply from recalling that Leith and Princes Street are heavily co-mentioned in Scottish literature – could reuse the same answer on the other two designs without engaging with either rendering at all, which would test recall of a previous answer rather than each design's own capacity to communicate connection strength. Item 2 cannot be scored automatically, but a participant's justification is expected to reference that specific design's visual encoding (edge thickness or node proximity for the force-directed network, arc height for the linear diagram, shared-line adjacency for the metro map); an answer that is generic, absent, or inconsistent with the encoding actually used is treated during analysis as evidence the corresponding item 1 answer may have been carried over rather than freshly determined. Items 4 and 7 are also now varied across the three topology designs rather than reusing identical place pairs and cluster options: for the force-directed network, the three rated pairs are New Town–Princes Street (weight 18), Lochend–Waverley Station (weight 4), and Leith–Silvermills (weight 2), and the verified cluster is New Town, Princes Street, Dundas Street, and Stockbridge (one Louvain community on the shared co-occurrence graph); for the linear diagram, the pairs are Dundas Street–Princes Street (17), Old Town–Princes Street (6), and Arthur's Seat–Haddington (2), and the verified cluster is Leith Walk, Lochend, Pilrig, and Waverley Station (a second, distinct Louvain community on the same graph); for the metro map, the pairs are Bruntsfield–Dundas Street (16), Howe Street–Stockbridge (6), and Dalkeith–Linlithgow (2), and the verified cluster is drawn from that design's own author-named lines (the stations of "Lockhart's Edinburgh"). A participant therefore has to locate a new set of places in whichever rendering is currently in front of them rather than reuse a rating or selection made on an earlier design.

Cross-design synthesis (shown once, after all three topology designs have been seen).

1. Having now seen all three ways of showing connections between places, is there anything about Edinburgh's literary geography that surprised you, or that you had not thought about before — for example, places that seem to "belong together" in these authors' writing that you would not have expected? *(free text, optional but encouraged)*

Post-task ranking (once per direction, after all three designs in that direction have been seen).

1. Of the three fingerprint designs you just saw (radar chart, bar-code, small multiples), which did you find the most intuitive to read without any explanation? *(single choice + free-text reason, optional)*

2. Of the three topology designs you just saw (force-directed network, linear connection diagram, metro map), which did you find the most intuitive to read without any explanation? *(single choice + free-text reason, optional)*

A.4 Survey Questions

Background (before the tasks). Collected to test whether familiarity moderates the results reported in Chapter 6, not for identification purposes, consistent with the ethics application's data-minimisation commitment.

1. How familiar are you with Edinburgh's literary history (e.g. Scott, Stevenson, contemporary Edinburgh fiction)? *(Not at all – Very familiar, 5-point scale)*

2. Have you read any works by Alexander McCall Smith, Irvine Welsh, John Gibson Lockhart, Walter Scott, or Robert Louis Stevenson? *(Yes, several / Yes, one or two / No)*

3. How would you describe your background? *(Literary studies / digital humanities / information visualisation or HCI / other academic / general public with no specialist background / other, please specify)* — this question, not a fixed group assignment, is what actually separates the "domain expert" and "general public" groups described in Chapter 6 at analysis time.

4. How familiar are you with reading data visualisations or charts in general? *(Not at all – Very familiar, 5-point scale)*

5. Have you lived in or spent significant time in Edinburgh? *(Yes, I live/have lived there / I've visited several times / I've visited once or twice / Never)* — included to separate familiarity with the city itself from familiarity with its literature, which the corpus analysis (Chapter 4) treats as distinct.

Reflection (after all tasks).

1. Before this study, had you thought about literary place names as anything other than locations on a map?
2. Did any of the six visualisations change how you think about the relationship between a literary work and the real city it is set in? If so, which one, and how?
3. Is there anything about any of the six visualisations that confused you or that you would change?
4. Any other comments?
5. Overall, which single visualisation of all six did you find the most memorable or interesting, and why? *(free text)*

Appendix B

Data and Implementation

B.1 Database Schema

The LitLong Edinburgh PostgreSQL database contains 27 tables after full import. The tables most relevant to this project are listed in Table B.1.

Table B.1: Key tables in the LitLong Edinburgh database.

Table	Rows	Description
`api_document`	620	Literary works (title, publication date, type)
`api_author`	424	Authors (name, gender, Open Library ID)
`api_document_author`	1,232	Many-to-many: documents ↔ authors
`api_location`	2,135	Place names (text, latitude, longitude)
`api_locationmention`	50,248	Mention records (start_word, page_id, sentence_id)
`api_sentence`	50,248	Full source text for each mention
`api_genre`	29	Literary genre classifications
`api_posmention`	2,085,834	Part-of-speech annotations
`mention_order`	50,248	Custom: narrative order and position (derived)

B.2 Sector Definitions

Table B.2 lists the 14 sectors used in the spatial classification, their data sources, and their place-name counts.

Note: Nearest-neighbour counts are included in the exact-match totals above. Source: City of Edinburgh Council Open Spatial Data Portal (Open Government Licence v3.0). "Copyright

Table B.2: The 14 spatial sectors and their place-name counts.

Sector	Source	Exact matches	Nearest-neighbour
Old Town	Natural Neighbourhoods	296	—
South Central	NP Areas	255	—
New Town	Natural Neighbourhoods	217	—
Leith	NP Areas	152	—
Inverleith	NP Areas	140	—
Craigentinny/Duddingston	NP Areas	73	—
Pentlands	NP Areas	66	—
Canongate	Natural Neighbourhoods	64	—
Almond	NP Areas	64	—
South West	NP Areas	44	—
Western Edinburgh	NP Areas	43	—
Portobello/Craigmillar	NP Areas	38	—
Liberton/Gilmerton	NP Areas	36	—
Forth	NP Areas	27	—
Outer Scotland (excluded)	—	232	—

City of Edinburgh Council, contains Ordnance Survey data © Crown copyright.”

B.3 Author Selection

The five test authors were selected on the basis of mention count, number of works in the corpus, and thematic contrast. Table B.3 summarises their corpus statistics and dominant sectors.

Table B.3: Five test authors and their corpus statistics.

Author	Mentions	Works	Dominant sector	%
Alexander McCall Smith	7,320	17	New Town	43.6
Irvine Welsh	2,634	8	Leith	44.3
John Gibson Lockhart	3,046	6	New Town	25.9
Walter Scott	2,796	12	Old Town	32.0
Robert Louis Stevenson	1,778	16	Old Town	20.7

B.4 Co-occurrence Analysis

Table B.4 lists the ten strongest document-level co-occurrence pairs in the corpus, computed across the five test authors.

Table B.4: Top 10 document-level co-occurrence pairs (five test authors).

Place A	Place B	Weight (documents)
Leith	Princes Street	28
New Town	Princes Street	18
Dundas Street	Princes Street	17
Bruntsfield	Dundas Street	16
Bruntsfield	Princes Street	16
Dundas Street	New Town	16
Dundas Street	Morningside	15
Morningside	Stockbridge	15
Princes Street	Stockbridge	15
Dundas Street	The Meadows	15

B.5 Metro-Map Line Composition

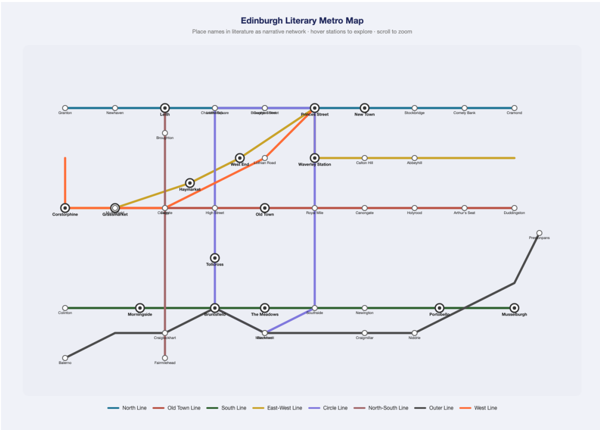

Figure B.1: The initial hand-drawn metro-map prototype, before the redesign described in Section 4.6. Lines follow geographic and cultural axes (a North Line, an Old Town Line, an East-West Line) chosen by inspection, with only 22% of adjacent station pairs backed by measured co-occurrence, against 88% for the redesigned map (Figure 4.6).

Table B.5 lists the five lines produced by the community-detection pipeline described in Section 4.6, with the number of home stations (stations whose narrative cluster this line is), the total station count including spliced-in interchange stops, total mentions summed across home stations, and the adjacency-backed diagnostic (the number of home-station-adjacent pairs, out of the total possible, that share at least one book in common).

Table B.5: The five metro-map lines and their composition.

Line	Home stations	Total stops	Mentions	Adjacency backed
Smith's Edinburgh	15	16	2,786	13/14 (93%)
Leith & Forth Corridor	12	23	1,564	9/11 (82%)
Scott's Edinburgh	13	14	826	9/12 (75%)
Welsh's Edinburgh	12	12	494	11/11 (100%)
Lockhart's Edinburgh	5	5	202	4/4 (100%)
Total	57	—	5,872	46/52 (88%)

Note: "Total stops" exceeds "home stations" where a line has been joined by an interchange station whose home cluster is a different line (12 stations are interchanges across the whole map). The 22% adjacency-backed figure for the earlier, hand-drawn eight-line version was computed over 67 same-line adjacent pairs, 15 of which shared a book.

B.6 Scale Exploration Summary

All six designs were re-tested at four benchmark author counts (2, 5, 20, and either 408 or the largest value practical for the design in question), plus a 2–3 book test for network and linear, using scale-exploration tools kept separate from the six canonical five-author prototypes (`data/processed/scale_exploration/`). Table B.6 summarises the failure mode identified for each design; the full quantitative detail (per-N tables, timing measurements, and the reasoning behind every scope choice) is recorded in the accompanying scale-exploration records submitted alongside this dissertation, rather than reproduced in full here.

Table B.6: Scaling failure mode identified for each of the six designs.

Design	Failure mode	Key evidence
Radar	Visual overlap	Polygons indistinguishable by N=20; unreadable at 408; hover-to-isolate recovers single-author legibility at any N.
Bar-code	Navigation cost	408 rows render in 20ms (no rendering problem) but the page is ≈45,000px tall, about 50 screens of scrolling.
Network	Visual overlap (graceful)	Core becomes a dense mass at 408 authors; periphery stays legible. Density is U-shaped in N (79.6% at 2 authors, ≈25% at 5, 16.5% at 408), driven by vocabulary non-overlap (Finding 1), not raw node count.
Linear	Horizontal sprawl	Canvas width grows from 1,100px (5 authors) to 29,790px (408 authors); a screenshot of the scrolled viewport looks unchanged because the same high-mention places sort leftmost regardless of N.
Small multiples	Computational cost (the only one)	Live Leaflet map initialisation degrades 3.6× per-panel from 50 to 408 authors (1.47s total at 408, measured directly); the one design where scale risks a real freeze, not just illegibility.
Metro	Clustering algorithm breakdown	Adjacency-backed proportion falls monotonically (96%→72%, N=2 to 50); at N=50 one community reaches 75 stations because modularity detection judges it unsplittable despite exceeding the 18-station cap.